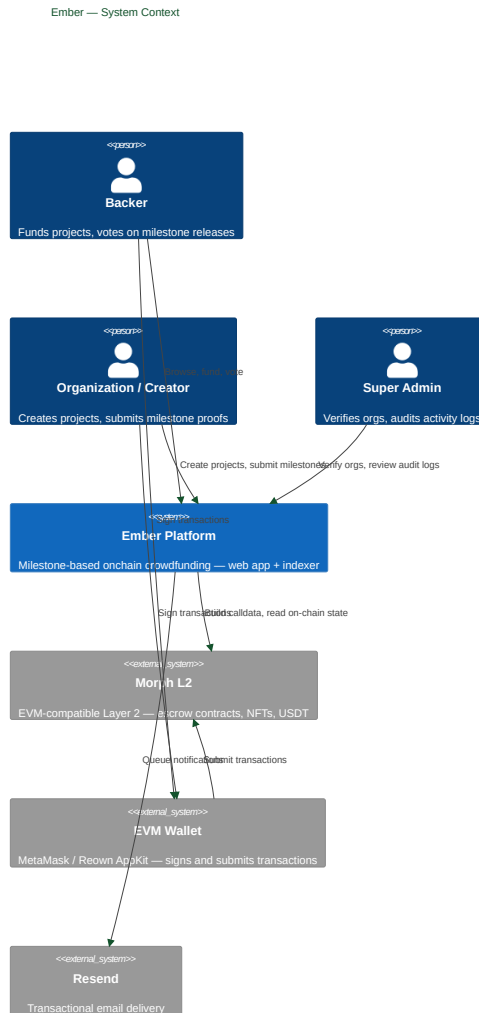


Ember — System Architecture

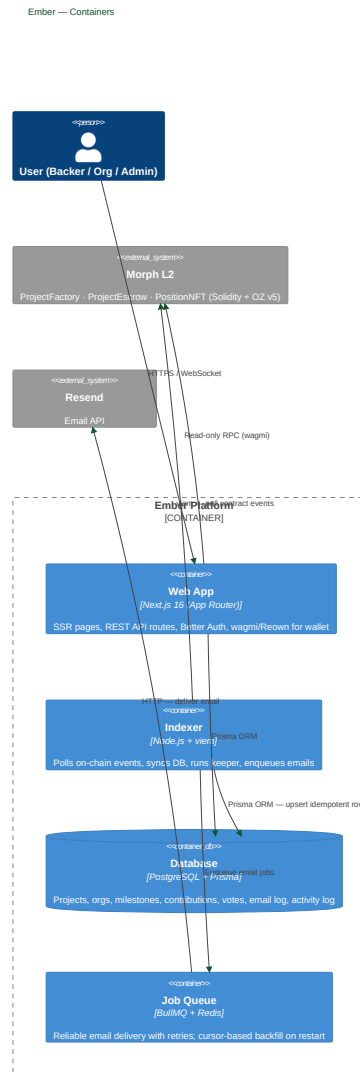
C4 — System Context

The actors around Ember and the external systems it integrates with. Ember's web app and indexer build calldata and read on-chain state; the user's wallet signs and submits every transaction to Morph L2.



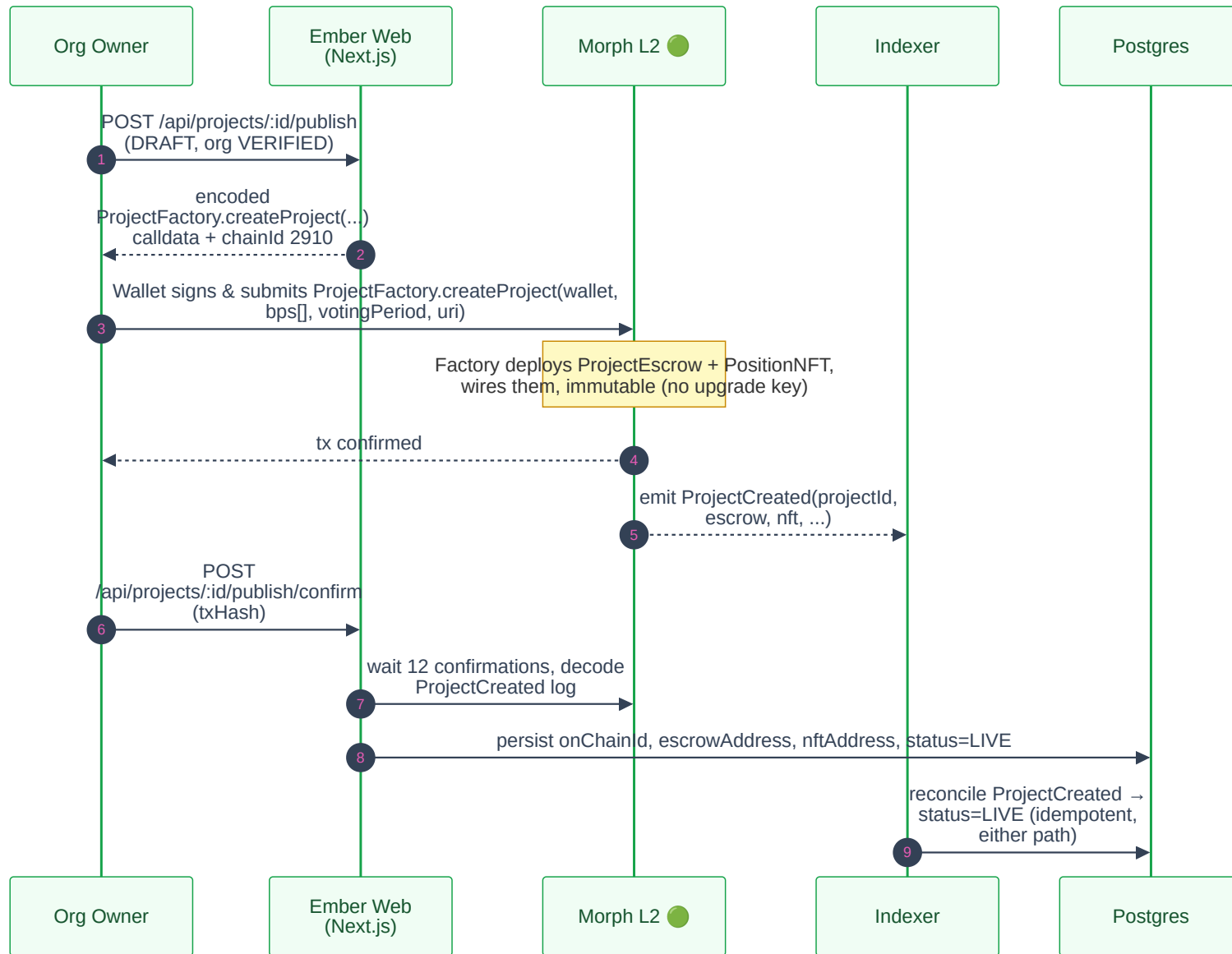
C4 — Container Diagram

Ember's runtime containers. The web app serves SSR pages and REST APIs; the indexer watches Morph events and syncs Postgres; BullMQ + Redis drive reliable email delivery.



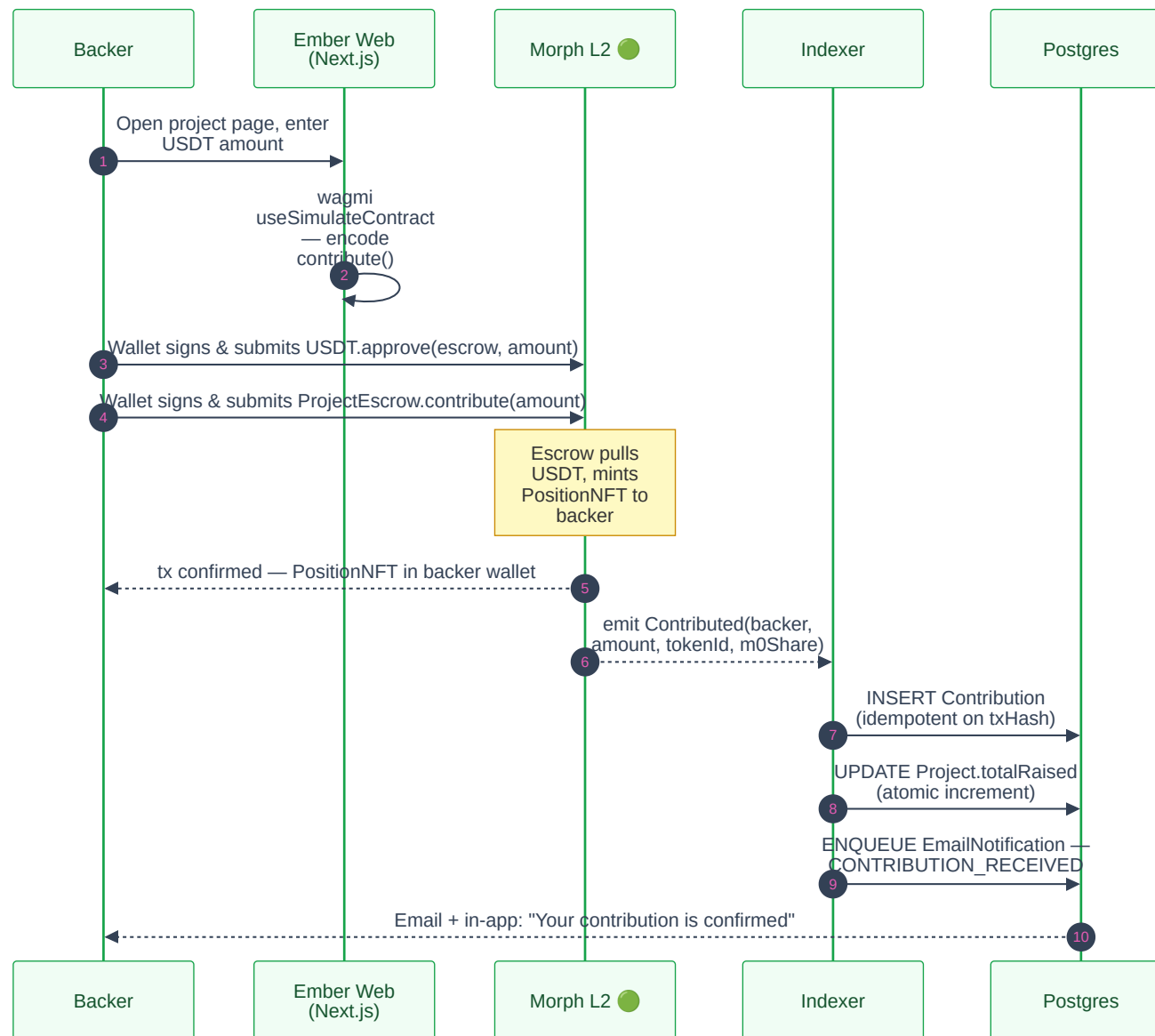
Sequence 1 — Project Creation

An org owner publishes a DRAFT project. The ProjectFactory contract on Morph deploys a dedicated ProjectEscrow + PositionNFT pair and emits ProjectCreated.



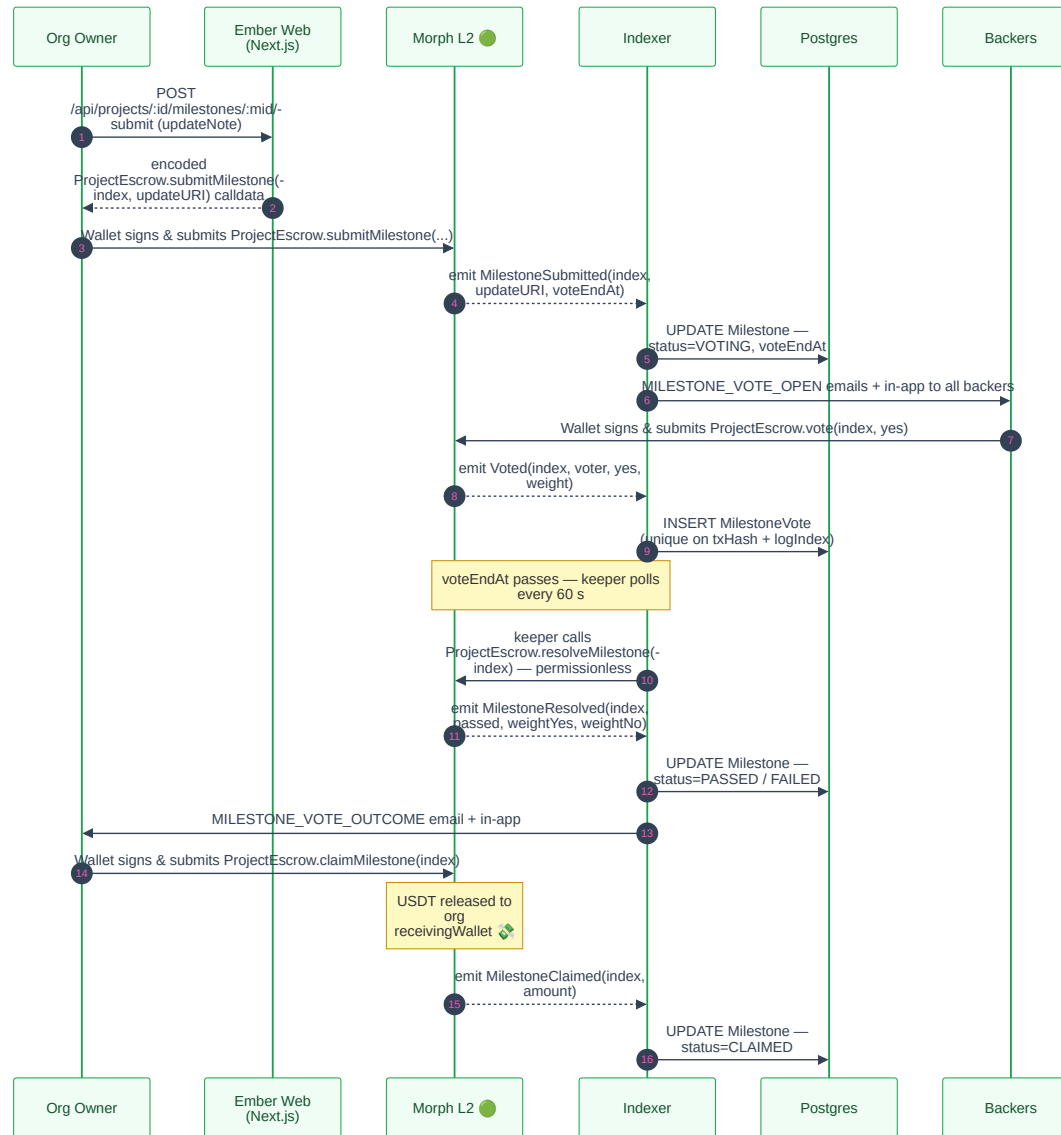
Sequence 2 — Backer Contributes to a Project

The backer approves USDT and calls `contribute()` on the project's escrow; the escrow pulls the USDT and mints a PositionNFT to the backer's wallet.



Sequence 3 — Milestone Submit → Vote → Release

The org submits proof for a milestone, backers vote weighted by their contribution, the permissionless keeper resolves the milestone after the window closes, and on a PASSED result the org claims its USDT.



Smart Contract Architecture (Morph L2)

Every project gets its own immutable ProjectEscrow + PositionNFT pair deployed by a singleton ProjectFactory. Contracts have no upgrade key — once deployed, the rules are fixed. Built with Solidity ^0.8.28 and OpenZeppelin v5.

```
ProjectFactory (singleton)
├─ createProject(orgWallet, milestoneBps[], votingPeriod, projectURI)
│   └─ deploys → ProjectEscrow (one per project, immutable, no upgrade key)
│       ├── contribute(amount)          USDT in → PositionNFT minted
│       ├── submitMilestone(index, uri)  org only
│       ├── vote(index, yes)             backers only, weighted by stake
│       ├── resolveMilestone(index)      permissionless, after voteEndAt
│       ├── claimMilestone(index)        org only, after PASSED → USDT out
│       └─ votingPowerOf(address)        view – weight = USDT contributed
│   └─ deploys → PositionNFT (ERC-721)
│       └─ tokenURI → /api/nft/:contract/:tokenId
└─ emits ProjectCreated(projectId, organization, creator, escrow, nft, bps[], votingPeriod)
    └─ indexed by Ember Indexer → sets Project status=LIVE
```

Events (mirrored into Postgres by the indexer): `ProjectCreated` , `Contributed` , `MilestoneSubmitted` , `Voted` , `MilestoneResolved` , `MilestoneClaimed` .

Contract	Responsibility	Key trait
ProjectFactory	Deploys and wires each project's escrow + NFT; assigns projectId	Singleton, immutable
ProjectEscrow	Holds USDT, runs contributions / weighted voting / milestone release	One per project, no upgrade key, nonReentrant on value paths
PositionNFT	ERC-721 minted to backers as transferable proof of stake	Escrow wired once via initEscrow